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 Instructor: Prof. A.K. Nandakumaran
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Foundations of Distribution Theory

These notes summarize core definitions and structural results concerning test functions, distributions, support, order, and distributional convergence.

1 Test Functions

Definition 1.1 ($\mathcal{D}(\Omega)$). Let $\Omega \subset \mathbb{R}^n$, open, then:

$$\mathcal{D}(\Omega) = C_c^\infty(\Omega) = \{\phi : \Omega \rightarrow \mathbb{R} \mid \phi \text{ is smooth and } \text{supp}(\phi) \subset\subset \Omega\}$$

So, $\mathcal{D}(\Omega)$ is a space of smooth functions or test functions, where $\phi \in C^\infty(\Omega)$ with a compact support. All these test functions are non-zero within the support and vanishes smoothly outside the support.

Definition 1.2 ($\mathcal{D}_K(\Omega)$). Let $\Omega \subset \mathbb{R}^n$ and $K \subset\subset \Omega$ compact. Then

$$\mathcal{D}_K(\Omega) = \{\phi \in C_c^\infty(\Omega) : \text{supp}(\phi) \subset K\}$$

The relationship between $\mathcal{D}(\Omega)$ and $\mathcal{D}_K(\Omega)$ is:

$$\mathcal{D}(\Omega) = \bigcup_{K \subset\subset \Omega} \mathcal{D}_K(\Omega)$$

Definition 1.3 (Seminorms). Seminorm is defined on $\mathcal{D}_K(\Omega)$ as

$$p_{K,m}(\phi) = \sum_{|\alpha| \leq m} \sup_{x \in K} |\partial^\alpha \phi(x)|$$

Remark 1.4. $\mathcal{D}_K(\Omega)$ endowed with locally convex topology generated by the family of seminorms $\{p_{K,m}\}_{m \in \mathbb{N}}$

1.1 Convergence in $\mathcal{D}(\Omega)$

A sequence $\{\phi_j\} \subset \mathcal{D}(\Omega)$, converges to 0 in $\mathcal{D}(\Omega)$ if there exists a compact set $K \subset\subset \Omega$ such that

1. $\text{supp}(\phi_j) \subset K$ for all j ,
2. for every multiindex α ,

$$\sup_{x \in K} |\partial^\alpha \phi_j(x)| \rightarrow 0 \quad \text{as } j \rightarrow \infty.$$

More generally, a sequence ϕ_j converges to ϕ in $\mathcal{D}(\Omega)$, if $\phi - \phi_j \rightarrow 0$ in the above sense.

2 Distributions

Definition 2.1 (Distribution via Sequential Continuity). Let $\Omega \subset \mathbb{R}^n$, open. A distribution on Ω is a linear functional

$$T : \mathcal{D}(\Omega) \rightarrow \mathbb{R}$$

such that whenever a sequence of test functions $\{\phi_j\} \subset \mathcal{D}(\Omega)$ converges to 0 in $\mathcal{D}(\Omega)$, we have

$$\langle T, \phi_j \rangle \rightarrow 0 \quad \text{as } j \rightarrow \infty$$

A distribution is a sequentially continuous linear functionals on $\mathcal{D}(\Omega)$.

Theorem 2.2 (Sequential Continuity Equivalent to Seminorm Estimate). A linear functional $T : \mathcal{D}(\Omega) \rightarrow \mathbb{R}$ on $\mathcal{D}(\Omega)$ is a distribution if and only if for every compact set $K \subset\subset \Omega$, there exists a constant $C > 0$ and $m \in \mathbb{N}$, such that

$$|\langle T, \phi \rangle| \leq Cp_{K,m}(\phi)$$

Hence, sequential continuity is equivalent to seminorm estimates.

Definition 2.3 (Order of a Distribution). Let $T \in \mathcal{D}'(\Omega)$ is a distribution of order $\leq m$ on a compact set $K \subset\subset \Omega$, if there exists a constant $C > 0$ for all multiindex α such that

$$|\langle T, \phi \rangle| \leq C \sum_{|\alpha| \leq m} \sup_{x \in K} |\partial^\alpha \phi(x)| \quad \forall \phi \in \mathcal{D}_K(\Omega)$$

If such an integer m can be chosen uniformly for all compact set K , then the distribution T will have finite order, and if not, then the distribution will have infinite order. Hence, order of a distribution depends on the compact set K . The seminorm estimate, for $m = 0$, reduces to

$$|\langle T, \phi(x) \rangle| \leq C \sup_{x \in K} |\phi|$$

This defines a distribution of order 0. For example, if the distribution is δ , then the seminorm estimate becomes:

$$|\delta(\phi)| = |\phi(0)| \leq \sup_{x \in K} |\phi(x)|$$

Thus, Dirac delta generates a zero-order distribution.

Remark 2.4. Distribution order is defined using seminorm estimates. Physically, order of a distribution says how many derivatives needed to control. Thus, more localized a distribution induced by locally integrable function, less derivatives can control the distribution.

3 Locally Integrable Functions as Distributions

Definition 3.1 (T_f). Let $\Omega \subset \mathbb{R}^n$, open and let function $f \in L^1_{\text{loc}}(\Omega)$. $K \subset\subset \Omega$ be a compact set. We define the linear functional $T_f : \mathcal{D}(\Omega) \rightarrow \mathbb{R}$ by

$$T_f(\phi) = \langle f, \phi \rangle = \int_{\Omega} f(x)\phi(x) dx, \quad \phi \in \mathcal{D}(\Omega)$$

Proposition 3.2. If $f \in L^1_{\text{loc}}(\Omega)$, then $T_f \in \mathcal{D}'(\Omega)$.

Proof. Let $\phi \in \mathcal{D}(\Omega)$ and set $K = \text{supp}(\phi) \subset\subset \Omega$. Since $f \in L^1_{\text{loc}}(\Omega)$, we have

$$\int_K |f(x)| dx < \infty.$$

Hence

$$|T_f(\phi)| = \left| \int_K f(x)\phi(x) dx \right| \leq \sup_{x \in K} |\phi(x)| \int_K |f(x)| dx,$$

which shows that $T_f(\phi)$ is well defined.

Now let $\{\phi_j\} \subset \mathcal{D}(\Omega)$ converge to 0 in $\mathcal{D}(\Omega)$. Then there exists a compact set K containing $\text{supp}(\phi_j)$ for all j , and

$$\sup_{x \in K} |\phi_j(x)| \rightarrow 0.$$

Therefore,

$$|T_f(\phi_j)| \leq \left(\int_K |f(x)| dx \right) \sup_{x \in K} |\phi_j(x)| \rightarrow 0,$$

which proves sequential continuity.

Linearity follows immediately from linearity of the integral. Let $\phi, \psi \in C_c^\infty(\Omega)$ and $\text{supp}(\phi) \subset K$, $\text{supp}(\psi) \subset K$. Then

$$T_f(a\phi + b\psi) = aT_f(\phi) + bT_f(\psi)$$

For $a, b \in \mathbb{R}$. The above equality holds true because $\langle f, \phi \rangle$ and $\langle f, \psi \rangle$ are well defined by sup bound. Thus, T_f follows linearity. Hence, T_f defines a distribution induced by locally integrable function f . $\square$

Remark 3.3. The above proof for T_f as a distribution, we only needed $m = 0$, which means for constant $C_K > 0$:

$$|T_f(\phi)| \leq C_K p_{K,0}(\phi)$$

Hence, T_f is of order zero, or more generally, any distribution induced by locally integrable function has order zero. This extends the previous remark that δ (and more generally, any distribution induced by a locally integrable function) has order zero.